

SDV_Replica_POC

Daily Work Log - Build Console Fix Session

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Project: SDV_Replica_POC

Log version: v3.1

Follow-up session: build_console.bat's first real run on Windows (all 6 modules) was shared back with logs showing every module failing. Diagnosed from the actual log files, fixed, re-verified build correctness (not the .bat script itself, which still needs a real Windows re-run), and added colour output as requested.

1. FIRST-RUN FAILURE - DIAGNOSIS

Item	Result
All 6 modules reported BUILD FAILED on first real Windows run	FAIL
All 6 log files read - every one showed the identical single line:	INFO
'cmake' is not recognized as an internal or external command	FAIL
Cross-checked: buildenv.sh never uses cmake (calls arm-none-eabi-gcc directly)	INFO
Conclusion: CMake was never installed on this machine at all	INFO
Separately noticed: 'Source count: 0 file(s)' on every module, including ones with real files	FAIL

In plain terms: This was not a bug in the repo, the C++ code, or the ZCU firmware - every module's log showed the exact same one-line failure, which only happens when the shell literally cannot find the cmake program at all. It never got far enough to even try reading a CMakeLists.txt file. Confirmed by checking that ALL SIX modules failed identically, and that the only Windows tool install done previously (the ARM toolchain) was never CMake.

2. FIXES APPLIED

- Added a pre-flight tool check (cmake / g++ / arm-none-eabi-gcc) that runs automatically before the menu appears - a missing tool is now reported in seconds, not discovered by reading 6 logs.
- Replaced the source-count for-loop with a dir+find based count - a known more reliable idiom.
- Added ANSI colour output using VT100 escape codes (native on Windows 10 1511+/11, no setup needed): magenta/yellow banners and headers, cyan in-progress, bright green PASS, bright red FAIL - deliberately different from the cyan/green reference example this was styled after.
- Re-verified: all 6 modules still compile and link correctly with the updated repo (g++/cmake, sandbox re-check) - the fixes did not touch or break any actual source code.

ACTION NEEDED - install CMake

Run in PowerShell: `winget install --id Kitware.CMake -e` - then close and reopen the terminal (same PATH-refresh requirement as the earlier ARM toolchain install) before re-running `build_console.bat`. The new pre-flight check will confirm cmake is found before you even reach the menu.

3. STILL OPEN

- `build_console.bat` still needs its first fully successful end-to-end run on real Windows.
- g++ availability on this machine is still unconfirmed - the pre-flight check will reveal this on the next run; report back if it shows MISSING.
- All other previously-open items (STM32 part numbers, linker script values, RPi5 bring-up, gRPC/protobuf logic) remain open.

End of log - 2026-08-30. Repo: github.com/SJLNS/SDV_Replica_POC